

BEND, OR, USAWMO: **720638**Lat: **44.095N**Lon: **121.200W**Elev: **3462**StdP: **12.95**Time Zone: **-8.00 (NAP)**Period: **09-19**WBAN: **00224****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
12	5.5	11.7	-1.7	5.7	21.7	1.4	6.7	24.9	27.3	46.0	24.4	45.0	3.4	340	0.576

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	32.7	92.5	59.6	89.9	59.2	85.9	58.4	62.3	84.4	60.9	83.6	59.6	81.6	8.7	0

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
54.7	72.5	67.2	53.6	69.6	65.9	50.3	61.5	64.8	29.6	84.0	28.6	84.7	27.6	82.2	68.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.5	16.4	DB	-3.5	97.3	8.5	2.9	-9.6	99.5	-14.5	101.2	-19.3	102.8	-25.5	105.0	(4)
(5)			WB	-4.2	65.7	8.2	1.9	-10.1	67.0	-14.9	68.1	-19.5	69.1	-25.5	70.5	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	48.7	34.6	35.7	40.6	44.9	53.1	59.1	68.3	67.9	59.8	48.5	39.0	32.0	(6)
(7)		DBStd	14.58	8.73	8.40	6.84	7.23	7.73	7.31	6.50	6.37	7.68	7.11	9.35	9.26	(7)
(8)		HDD50	2458	478	402	297	190	56	10	0	0	11	113	342	559	(8)
(9)		HDD65	6295	942	820	756	604	374	205	44	43	190	513	781	1023	(9)
(10)		CDD50	1984	1	2	6	36	154	282	566	555	304	66	11	1	(10)
(11)		CDD65	345	0	0	0	0	6	27	145	133	33	1	0	0	(11)
(12)		CDH74	6236	0	0	0	33	205	613	2496	2110	733	46	0	0	(12)
(13)		CDH80	2500	0	0	0	3	37	180	1127	898	248	7	0	0	(13)
(14)	Wind	WSAvg	6.0	5.5	6.4	7.0	6.7	5.8	5.8	5.5	5.5	5.8	6.2	5.8	(14)	
(15)	Precipitation	PrecAvg	11.4	1.8	1.0	0.9	0.7	0.9	0.7	0.5	0.6	0.4	0.6	1.5	2.0	(15)
(16)		PrecMax	19.2	4.4	3.7	2.8	2.2	4.1	3.2	3.7	2.8	1.7	2.2	4.6	8.7	(16)
(17)		PrecMin	5.4	0.1	0.1	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.0	(17)
(18)		PrecStd	3.0	1.0	0.8	0.6	0.5	0.9	0.7	0.7	0.6	0.4	0.4	1.3	1.7	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	60.9	61.4	67.9	78.8	84.1	90.4	96.8	97.3	90.9	80.9	67.7	55.1	(19)
(20)			MCWB	44.4	46.9	47.3	53.4	56.9	62.1	61.0	60.4	57.9	53.6	49.3	44.2	(20)
(21)		2%	DB	54.2	55.5	63.3	72.0	79.4	84.3	92.8	91.5	86.1	72.8	61.3	52.0	(21)
(22)			MCWB	43.4	43.7	45.2	50.7	55.0	59.3	60.0	58.7	57.3	50.8	48.2	43.5	(22)
(23)		5%	DB	49.7	52.0	57.1	65.6	75.1	81.4	90.3	89.5	82.1	67.7	56.7	47.8	(23)
(24)			MCWB	41.3	42.1	43.6	48.4	53.1	56.8	59.3	58.5	56.1	49.6	45.9	41.7	(24)
(25)		10%	DB	46.1	48.1	54.0	59.4	71.5	77.0	86.3	84.4	77.3	63.1	52.5	45.2	(25)
(26)			MCWB	39.1	40.6	42.8	44.7	51.8	54.6	58.6	57.9	54.4	48.5	43.9	39.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	48.6	48.4	51.3	54.9	59.1	64.7	64.7	63.4	62.0	56.1	52.4	47.5	(27)
(28)			MCD B	56.9	59.0	62.0	72.5	76.7	84.0	87.3	86.2	83.5	72.4	62.4	53.6	(28)
(29)		2%	WB	44.2	45.9	47.9	51.4	56.9	61.4	62.3	61.7	59.7	53.3	50.1	44.3	(29)
(30)			MCD B	51.2	55.3	57.4	68.9	74.4	81.5	85.3	84.7	79.1	67.4	59.0	49.4	(30)
(31)		5%	WB	41.8	43.5	45.2	48.9	54.8	58.6	60.9	60.4	57.7	51.3	47.6	41.8	(31)
(32)			MCD B	48.5	49.9	54.9	63.0	70.2	77.0	84.9	82.8	78.3	63.9	56.8	47.1	(32)
(33)		10%	WB	39.6	40.9	43.2	46.4	52.9	56.3	59.5	59.1	56.0	49.4	44.1	39.2	(33)
(34)			MCD B	45.8	47.4	52.8	58.5	67.7	73.3	83.6	80.8	75.1	61.3	51.2	44.2	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	17.0	18.1	21.3	23.9	26.5	27.9	32.7	31.9	29.7	25.2	20.0	16.6	(35)
(36)			MCDBR	23.3	23.7	28.1	34.2	33.5	33.5	35.3	35.1	36.0	34.7	25.4	19.1	(36)
(37)			MCWBR	14.2	14.1	14.8	15.6	13.3	13.0	11.8	11.6	14.6	16.8	14.3	12.0	(37)
(38)		5% WB	MCDBR	20.4	20.7	25.2	30.9	28.9	30.8	33.0	32.4	32.6	27.5	23.1	16.7	(38)
(39)			MCWBR	13.4	13.0	14.5	15.1	12.1	12.9	11.8	11.3	13.5	14.4	13.4	11.6	(39)
(40)	Clear-Sky Solar Irradiance	taub		0.278	0.283	0.288	0.307	0.324	0.314	0.326	0.315	0.299	0.297	0.276	0.276	(40)
(41)		taud		2.543	2.561	2.571	2.512	2.487	2.543	2.518	2.565	2.585	2.579	2.597	2.544	(41)
(42)		Ebn at Noon		275	294	305	305	300	302	297	298	297	284	275	264	(42)
(43)		Edh at Noon		21	24	27	32	34	32	32	30	27	23	20	19	(43)
(44)	All-Sky Solar Radiation	RadAvg		505	790	1171	1636	1999	2318	2474	2128	1629	1018	581	409	(44)
(45)		RadStd		61	95	100	140	150	125	101	112	87	83	58	59	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)		
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (14 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+162	+63

Nomenclature: See separate page